

Chapter 12: Reactions Involving Pure Condensed Phases and a Gaseous Phase

Part I

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12.1. Introduction

- ✓ The criterion for equilibrium in a gaseous reaction system was discussed in Chapter 11. The question now is, how is the situation changed if one or more of the reactants or products of the reaction occurs as a condensed phase? In this chapter, we consider condensed phases that are of pure species—that is, are of fixed composition.
- ✓ Many practical systems occur in this category: for example, the reaction of pure metals with gaseous elements to form pure metal oxides, sulfides, halides, and so on.

12.1. Introduction

✓ Questions of interest include

1. What is the maximum oxygen pressure which can be tolerated in a gaseous atmosphere without oxidation of a given metal occurring at a given temperature?
2. To what temperature must a given carbonate be heated in a gaseous atmosphere of given partial pressure of carbon dioxide to cause the decomposition of the carbonate?

12.1. Introduction

- ✓ The first question is of interest in any of the many annealing processes that are performed in the laboratory or industrial plants. The second is of interest in, for example, the production of lime from limestone.
- ✓ In such systems, complete equilibrium entails the establishment of
 - Phase equilibrium among the individual condensed phases and the gas phase
 - Reaction equilibrium among the various species present in the gas phase

12.1. Introduction

- ✓ Since phase equilibrium is established when the pure condensed phases exert their saturated vapor pressures, which are uniquely fixed when the temperature of the system is fixed, then the only pressures which can be varied at constant temperature are those of the species which exist only in the gas phase.
- ✓ The unique variations, with temperature, of the saturated vapor pressures of pure condensed-phase species and the relative insensitivity of the Gibbs free energies of condensed phases to changes in pressure considerably simplify the thermodynamic treatment of reaction equilibria in systems containing both gaseous and pure condensed phases.

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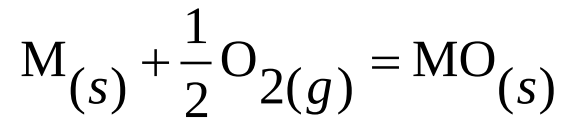
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12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ Consider the reaction equilibrium between a pure solid metal M, its pure oxide* MO, and oxygen gas at the temperature T and the pressure P :



- ✓ Here we assume that the oxygen is insoluble in the solid metal and that MO is stoichiometric.
- ✓ Both the metal M and the oxide MO are in equilibrium with their vapor species in the gas phase, as is required by the criterion for phase equilibrium; that is,

$$\bar{G}_{\text{M}}(\text{in the gas phase}) = G_{\text{M}}(\text{in the solid metal phase})$$

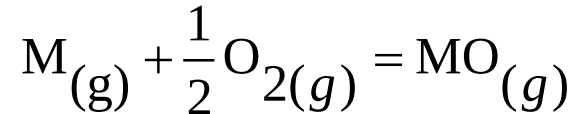
and

$$\bar{G}_{\text{MO}}(\text{in the gas phase}) = G_{\text{MO}}(\text{in the solid oxide phase})$$

* By *pure oxide* , it is meant that there is no third element dissolved in it. It acts as a single component.

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ Thus, the reaction equilibrium



is established in the gas phase.

- ✓ From Equation 11.6, the criterion for this reaction equilibrium to occur at the temperature T is

$$G_{\text{MO}(g)}^{\circ} - \frac{1}{2} G_{\text{O}_2(g)}^{\circ} - G_{\text{M}(g)}^{\circ} = -RT \ln \frac{p_{\text{MO}}}{p_{\text{M}} p_{\text{O}_2}^{1/2}} \quad (12.1)$$

or

$$\Delta G^{\circ} = -RT \ln \frac{p_{\text{MO}}}{p_{\text{M}} p_{\text{O}_2}^{1/2}}$$

where ΔG° is the difference between the Gibbs free energy of 1 mole of gaseous MO at 1 atm pressure and the sum of the Gibbs free energies of $\frac{1}{2}$ mole of oxygen gas at 1 atm pressure and 1 mole of gaseous M at 1 atm pressure, all at the temperature T .

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ Since M and MO are present in the system as pure solids, phase equilibrium requires that p_{MO} in Equation 12.1 be the saturated vapor pressure of solid MO at the temperature T , and that p_{M} be the saturated vapor pressure of solid M at the temperature T .
- ✓ Thus, the values of p_{MO} and p_{M} in the gas phase are uniquely fixed by the temperature T , and so the value of p_{O_2} in Equation 12.1, at which reaction equilibrium is established, is fixed at the temperature T .
- ✓ As stated previously, phase equilibrium in the system requires that

$$\bar{G}_{\text{M}}(\text{in the gas phase}) = G_{\text{M}}(\text{in the solid metal phase}) \quad (12.2)$$

and

$$\bar{G}_{\text{MO}}(\text{in the gas phase}) = G_{\text{MO}}(\text{in the solid oxide phase}) \quad (12.3)$$

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

✓ Equation 12.2 can be written as

$$G_{M(g)}^{\circ} + RT \ln p_{M(g)} = G_{M(s)}^{\circ} + \int_{P=1}^{P=p_{M(g)}} V_{M(s)} dP \quad (12.4)$$

and Equation 12.3 can be written as

$$G_{MO(g)}^{\circ} + RT \ln p_{MO(g)} = G_{MO(s)}^{\circ} + \int_{P=1}^{P=p_{MO(g)}} V_{MO} dP \quad (12.5)$$

✓ Consider Equation 12.4.

- $G_{M(s)}^{\circ}$ is the molar Gibbs free energy of solid M under a pressure of 1 atm at the temperature T .
- The integral $\int_{P=1}^{P=p_{M(g)}} V_{M(s)} dP$ is the effect of a change in pressure from $P = 1$ atm to $P = P$ on the value of the molar Gibbs free energy of solid M at the temperature T (where $V_{M(s)}$ is the molar volume of the solid metal at the pressure P and temperature T).

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ In the following, it is shown that the value of the integral is negligible compared with the other values in Equation 12.4, and hence Equation 12.4 can be written as

$$G_{M(g)}^{\circ} + RT \ln p_{M(g)} = G_{M(s)}^{\circ} \quad (12.4a)$$

- ✓ As an example to show this is the case, consider iron as a typical metal at a temperature of 1000 °C. The saturated vapor pressure of solid iron at 1000 °C is 6×10^{-10} atm, and thus, the term $RT \ln p_{M(g)}$ has the value $8.3144 \times 1273 \times \ln(6 \times 10^{-10}) = -224,750$ J. The molar volume of solid iron at 1000 °C is 7.34 cm^3 , which in the range 0–1 atm is independent of pressure. The value of the integral for $P = 6 \times 10^{-10}$ atm is $-7.34 \times 1 \text{ cm}^3 \cdot \text{atm} = -0.74$ J.

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ It is thus seen that $G_{\text{Fe}(g)}^\circ$ at 1000 °C is much larger in value than $G_{\text{Fe}(g)}^\circ$ at 1000 °C, which is to be expected in view of the large metastability of the iron vapor at 1 atm pressure and a temperature of 1000 °C with respect to the solid. Since the value of the integral (-0.74 J), is small compared with $-224,750$ J, Equation 12.4 can be written as Equation 12.4a.
- ✓ As a consequence of the negligible effect of pressure on the Gibbs free energy of a condensed phase (when the pressure is in the range 0–1 atm), the standard state of a species occurring as a condensed phase can be defined as being the *pure species at the temperature T* ; the specification that the pressure be 1 atm is no longer required. $G_{\text{M}(s)}^\circ$ is now simply the molar Gibbs free energy of pure solid M at the temperature T .

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

✓ Similarly, Equation 12.5 can be written as

$$G_{\text{MO}(g)}^{\circ} + RT \ln p_{\text{MO}(g)} = G_{\text{MO}(s)}^{\circ} \quad (12.5a)$$

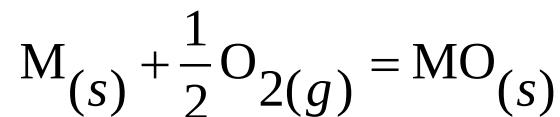
and thus, Equation 12.1 can be written as

$$G_{\text{MO}(s)}^{\circ} - \frac{1}{2} G_{\text{O}_2(g)}^{\circ} - G_{\text{M}(s)}^{\circ} = -RT \ln \left(\frac{1}{p_{\text{O}_2}^{1/2}} \right)$$

or

$$\Delta G^{\circ} = -RT \ln K \quad (12.6)$$

Where $K = 1 / p_{\text{O}_2}^{1/2}$, and ΔG° is the *standard Gibbs free energy change for the reaction*



12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ Thus, in the case of a reaction equilibrium involving only pure condensed phases and a gas phase, the equilibrium constant K can be written solely in terms of those species which occur only in the gas phase. Again, since ΔG° is a function only of temperature, then K is a function only of temperature.
- ✓ Thus, at any fixed temperature, the establishment of reaction equilibrium occurs at a unique value of $p_{\text{O}_2} = p_{\text{O}_2(\text{eq.}T)}$. The equilibrium has one thermodynamic degree of freedom, as can be seen from application of the Gibbs equilibrium phase rule:

$$\Phi = 3 \text{ (two pure solids and a gas phase)}$$

$$C = 2 \text{ (metal M+ oxygen)}$$

- ✓ Thus,

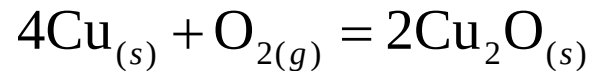
$$F = C + 2 - \Phi = 2 + 2 - 3 = 1$$

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ If, at any temperature T , the actual partial pressure of oxygen in a closed metal–metal oxide–oxygen system is greater than $p_{\text{O}_2(\text{eq.}T)}$, spontaneous oxidation of the metal will occur, consuming oxygen and decreasing the oxygen pressure in the gas phase.
- ✓ When the actual oxygen pressure has thus been decreased to $p_{\text{O}_2(\text{eq.}T)}$, then, provided that both solid phases are still present, the oxidation ceases and reaction equilibrium is reestablished.
- ✓ Similarly, if the oxygen pressure in the closed vessel was originally less than $p_{\text{O}_2(\text{eq.}T)}$, spontaneous reduction of the oxide would occur until $p_{\text{O}_2(\text{eq.}T)}$ was reached.

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ Extraction metallurgical processes involving the reduction of oxide ores depend on the achievement and maintenance of an oxygen pressure less than $p_{\text{O}_2(\text{eq.}T)}$ in the reaction vessel.
- ✓ For example, the standard Gibbs free energy change for the reaction



is

$$\Delta G^\circ = -324,400 + 138.5T \text{ J}$$

in the temperature range 298–1200 K. Thus,

$$-\ln K = \ln p_{\text{O}_2(\text{eq.}T)} = \frac{\Delta G^\circ}{RT}$$

or

$$\log p_{\text{O}_2(\text{eq.}T)} = -\frac{324,400}{2.303 \times 8.3144T} + \frac{138.5}{2.303 \times 8.3144} = -\frac{16,940}{T} + 7.23$$

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

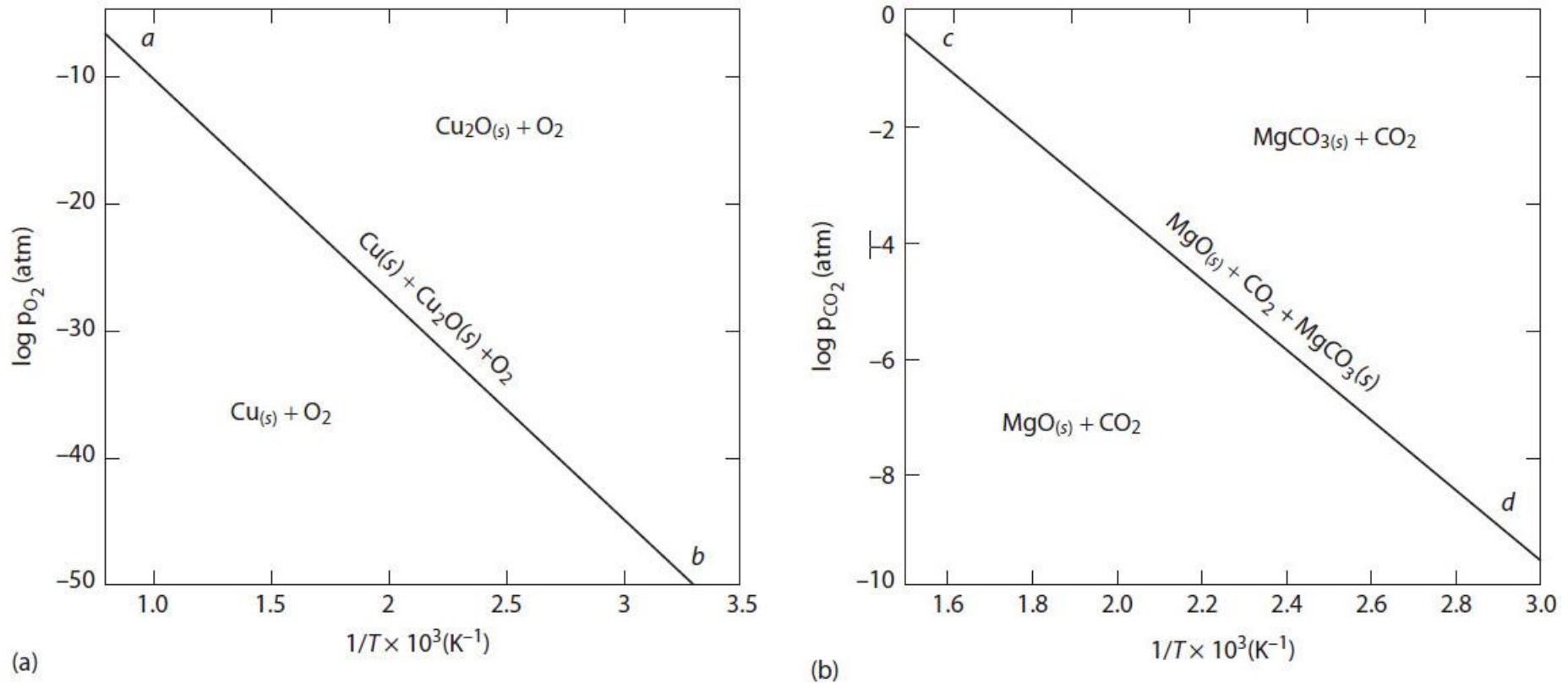


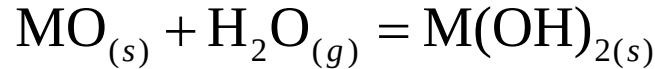
Figure 12.1 (a) The variation, with temperature, of the oxygen pressure required for maintenance of the equilibrium $4\text{Cu}_{(s)} + \text{O}_2 = 2\text{Cu}_2\text{O}$. (b) The variation, with temperature, of the carbon dioxide pressure required for maintenance of the equilibrium $\text{MgO}_{(s)} + \text{CO}_2 = \text{MgCO}_{3(s)}$.

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ This variation of $\log p_{\text{O}_2(\text{eq.}T)}$ with $1/T$ is drawn as the line *ab* in Figure 12.1a, and all points on the line represent the unique oxygen pressure, $p_{\text{O}_2(\text{eq.}T)}$ required for equilibrium between solid Cu, solid Cu_2O , and oxygen gas at the particular temperature T .
- ✓ Thus, the line *ab* divides the diagram into two regions. Above the line *ab* (where $p_{\text{O}_2} > p_{\text{O}_2(\text{eq.}T)}$), the metal phase is not stable, and thus, the system exists as $\text{Cu}_2\text{O}_{(s)} + \text{O}_{2(g)}$; below the line *ab* (where $p_{\text{O}_2} < p_{\text{O}_2(\text{eq.}T)}$), the oxide is not stable, and hence the system exists as $\text{Cu}_{(s)} + \text{O}_{2(g)}$.
- ✓ Other equilibria among two condensed pure phases and a gas phase include the formation of hydroxides and carbonates.

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

- ✓ For example, at the temperature T , the equilibrium



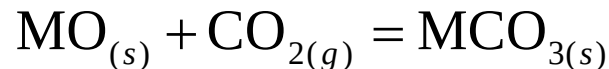
occurs when

$$G_{\text{MO}(s)}^{\circ} + G_{\text{H}_2\text{O}(g)}^{\circ} + RT \ln p_{\text{H}_2\text{O}} = G_{\text{M}(\text{OH})_2(s)}^{\circ}$$

that is, when

$$\Delta G^{\circ} = -RT \ln K = RT \ln p_{\text{H}_2\text{O}(\text{eq.}T)}$$

- ✓ Similarly, the equilibrium



occurs when

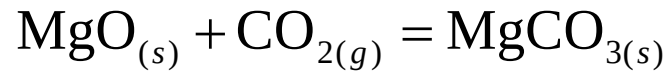
$$G_{\text{MO}(s)}^{\circ} + G_{\text{CO}_2(g)}^{\circ} + RT \ln p_{\text{CO}_2(\text{eq.}T)} = G_{\text{MCO}_3(s)}^{\circ}$$

that is, when

$$\Delta G^{\circ} = -RT \ln K = RT \ln p_{\text{CO}_2(\text{eq.}T)}$$

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

✓ For the reaction



$$\Delta G^\circ = -117,600 + 170T \text{ J}$$

in the temperature range 298 to 1000 K, and thus,

$$\log p_{\text{CO}_2(\text{eq.}T)} = -\frac{117,600}{2.303 \times 8.3144T} + \frac{170}{2.303 \times 8.3144} = -\frac{6141}{T} + 8.88$$

✓ This variation is shown in Figure 12.1b as the line *cd*, which again divides the diagram into two regions: one in which $\text{MgO}_{(s)} + \text{CO}_{2(g)}$ are stable, and one in which $\text{MgCO}_{3(s)} + \text{CO}_{2(g)}$ are stable.

12.2 Reaction Equilibrium In a System Containing Pure Condensed Phases and a Gas Phase

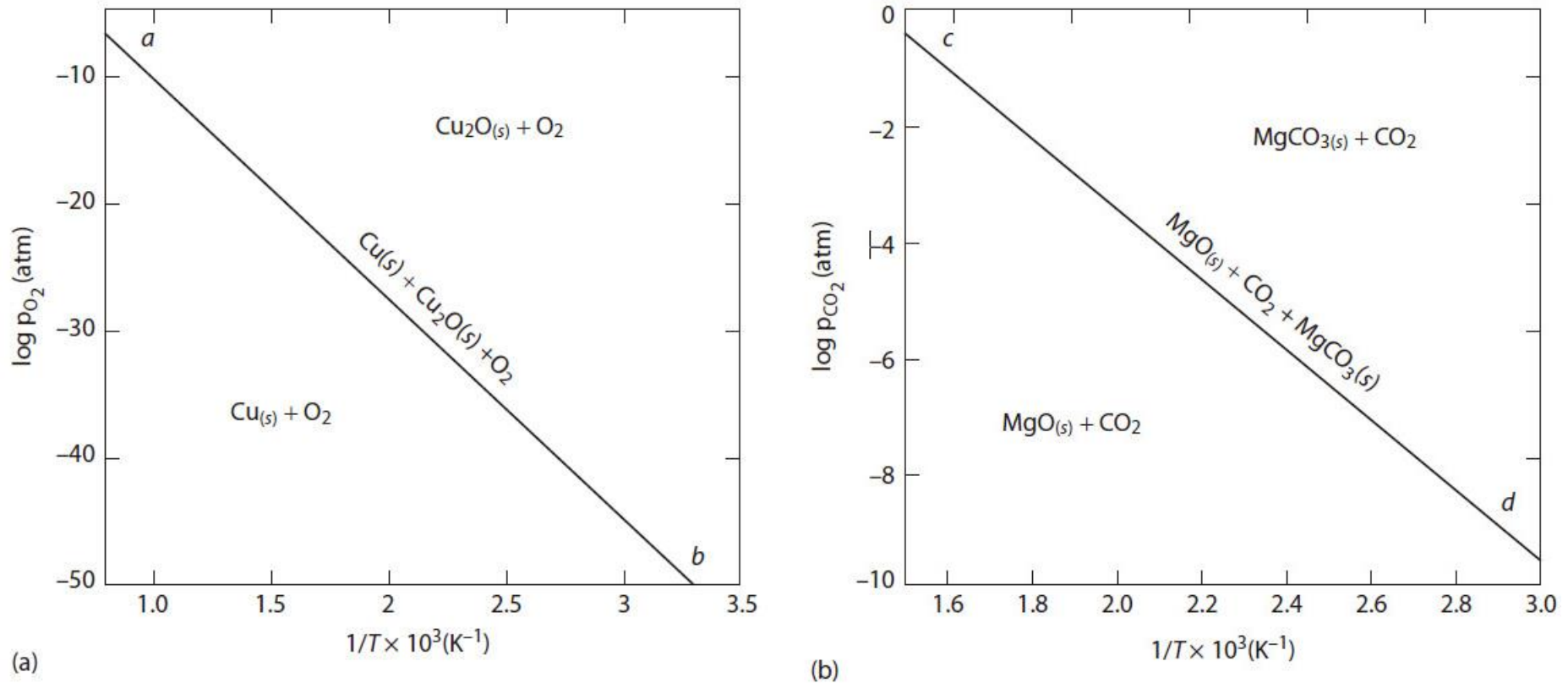


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12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ For any chemical reaction, combination of Equations 6.11 and 6.12 gives ΔG° for the reaction as a function of temperature:

$$\Delta G_T^\circ = \Delta H_T^\circ - T \Delta S_T^\circ = \Delta H_{298}^\circ + \int_{298}^T \Delta c_p dT - T \Delta S_{298}^\circ - T \int_{298}^T \frac{\Delta c_p}{T} dT \quad (12.7)$$

from which it is seen that the deviation from linearity between ΔG° and T depends on the sign and the magnitude of Δc_p for the reaction.

- ✓ Generally, however, the variation of ΔG° with T is considered as follows. For each of the individual reactants and products of the reaction, the molar heat capacity, c_p , is expressed, over a stated range of temperature, in the form

$$c_p = a + bT + cT^{-2}$$

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ Hence, for the reaction, again within the stated range of temperature,

$$\Delta c_p = \Delta a + \Delta bT + \Delta cT^{-2}$$

- ✓ From Kirchhoff's law:

$$\left(\frac{\partial \Delta H^\circ}{\partial T} \right)_p = \Delta c_p = \Delta a + \Delta bT + \Delta cT^{-2}$$

where ΔH° is the standard enthalpy change for the reaction.

- ✓ Integration gives

$$\Delta H_T^\circ = \Delta H_0 + \Delta aT + \frac{\Delta bT^2}{2} - \frac{\Delta c}{T} \quad (12.8)$$

where ΔH_0 is an integration constant which would be equal to the standard enthalpy of the reaction at 0 K only if the analytical expression for Δc_p as a function of T was valid down to 0 K. ΔH_0 is normally evaluated by substituting a known value of ΔH_T° into Equation 12.8.

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ The Gibbs–Helmholtz equation for the reaction is

$$\frac{\partial \left(\frac{\Delta G^\circ}{T} \right)}{\partial T} = -\frac{\Delta H^\circ}{T^2} = -\frac{\Delta H_0}{T^2} - \frac{\Delta a}{T} - \frac{\Delta b}{2} + \frac{\Delta c}{T^3}$$

integration of which gives

$$\frac{\Delta G^\circ}{T} = I + \frac{\Delta H_0}{T} - \Delta a \ln T - \frac{\Delta b T}{2} - \frac{\Delta c}{2T^2}$$

or

$$\Delta G^\circ = IT + \Delta H_0 - \Delta a T \ln T - \frac{\Delta b T^2}{2} - \frac{\Delta c}{2T} \quad (12.9)$$

where I is an integration constant.

- ✓ Since $\Delta G^\circ = -RT \ln K$, then Equation 12.9 gives

$$\ln K = -\frac{\Delta H_0}{RT} - \frac{I}{R} + \frac{\Delta a \ln T}{R} + \frac{\Delta b T}{2R} + \frac{\Delta c}{2RT^2} \quad (12.10)$$

- ✓ In Appendix 12A, the value of the constant I is found to be 171.7 J/K.

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ The variation of ΔG_T° , as calculated from the experimentally measured variation of $p_{\text{O}_2(\text{eq.}T)}$ with temperature, can be fitted to an equation of the form

$$\Delta G^\circ = A + BT \ln T + CT$$

- ✓ For the oxidation of $4\text{Cu}_{(s)}$ to $2\text{Cu}_2\text{O}_{(s)}$, this gives

$$\Delta G^\circ = -338,900 - 14.2T \ln T + 247T \text{ J} \quad (\text{ii})$$

which can be approximated in linear form by

$$\Delta G^\circ = -333,000 + 141.3T \text{ J} \quad (\text{iii})$$

Equations 12A (i), (ii), and (iii) give, respectively, -282.8 , -289.1 , and -290.6 kJ, and, at 1200 K, give -169.7 , -163.3 , and -163.4 kJ.

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ It can be noticed that Equation 12.10 is similar to the vapor pressure equation (Equation 7.8). The relationship between the two can be seen as follows. Consider the evaporation of A :

$$A_{(l)} = A_{(v)}$$

- ✓ Equilibrium occurs at the temperature T when

$$G_{A(l)}^{\circ} = G_{A(v)}^{\circ} + RT \ln p_A$$

that is, when

$$\Delta G^{\circ} = -RT \ln p_A = -RT \ln K$$

- ✓ If the liquid and the vapor have the same molar heat capacity, then

$$\ln p_A = -\frac{\Delta G^{\circ}}{RT} = -\frac{\Delta H^{\circ}}{RT} + \frac{\Delta S^{\circ}}{R}$$

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ This is to be compared with Equation 7.6a (Chapter 7), which gave

$$\ln p_A = -\frac{\Delta H_{\text{evap}}}{RT} + \text{constant}$$

- ✓ If the vapor behaves ideally, then, at constant temperature, $H_{(v)}$ is independent of pressure, and thus,

$$\Delta H^\circ = H_{(v)}^\circ - H_{(l)}^\circ = H_{(v)} - H_{(l)} = \Delta H_{\text{evap}}$$

- ✓ However, from Equation 6.15, for the vapor

$$S(T, p_A) = S_T^\circ - R \ln p_A$$

and thus,

$$\begin{aligned}\Delta S_{(l \rightarrow v)} &= S_{(v)}^\circ - S_{(l)}^\circ \\ &= S_{(v)} - S_{(l)}^\circ + R \ln p_A \\ &= \Delta S_{\text{evap}} + R \ln p_A\end{aligned}$$

12.3 The Variation of the Standard Gibbs Free Energy Change with Temperature

- ✓ The constant in Equation 7.6a thus has the value $\Delta S_{\text{evap}}/R + \ln p_A$, and Equation 7.6a becomes

$$\ln p_A = -\frac{\Delta H_{\text{evap}}}{RT} + \frac{\Delta S_{\text{evap}}}{R} + \ln p_A$$

indicating that when the vapor is in equilibrium with the liquid at the temperature T ,

$$\Delta S_{\text{evap}} = \frac{\Delta H_{\text{evap}}}{T}$$

or, as is required,

$$\Delta G_{\text{evap}} = 0$$

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12.4 Ellingham Diagrams

- ✓ Ellingham* (Harold Johann Thomas Ellingham, 1897– 1975) plotted the experimentally determined variations of ΔG° with T for the oxidation and sulfidation of a series of metals and found that, in spite of the terms involving $\ln T$, T^2 , and T^{-1} in Equation 12.9, the relationship approximated to straight lines over ranges of temperature in which no change of state occurred.

* H. J. T. Ellingham, “ Reducibility of Oxides and Sulfides in Metallurgical Processes,” J. Soc. Chem. Ind. (1944), vol. 63, p. 125.

12.4 Ellingham Diagrams

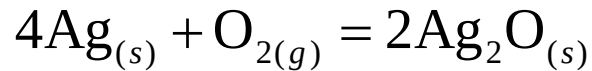
- ✓ The relations could thus be expressed by means of the simple equation

$$\Delta G^{\circ} = A + BT \quad (12.11)$$

in which the constant A is identified with the temperature-independent standard enthalpy change for the reaction, ΔH° , and the constant B is identified with the negative of the temperature-independent standard entropy change for the reaction, $-\Delta S^{\circ}$.

12.4 Ellingham Diagrams

- ✓ The variations of ΔG° with T for the oxidation reaction



is shown in Figure 12.2, which is known as an *Ellingham diagram*.

- ✓ From Equation 12.11, the length of the intercept of the line with the $T = 0$ K axis gives ΔH° , and ΔS° is given by the negative of the slope of the line.
- ✓ Since ΔS° is a negative quantity (the reaction involves the disappearance of a mole of gas), the line has a positive slope.

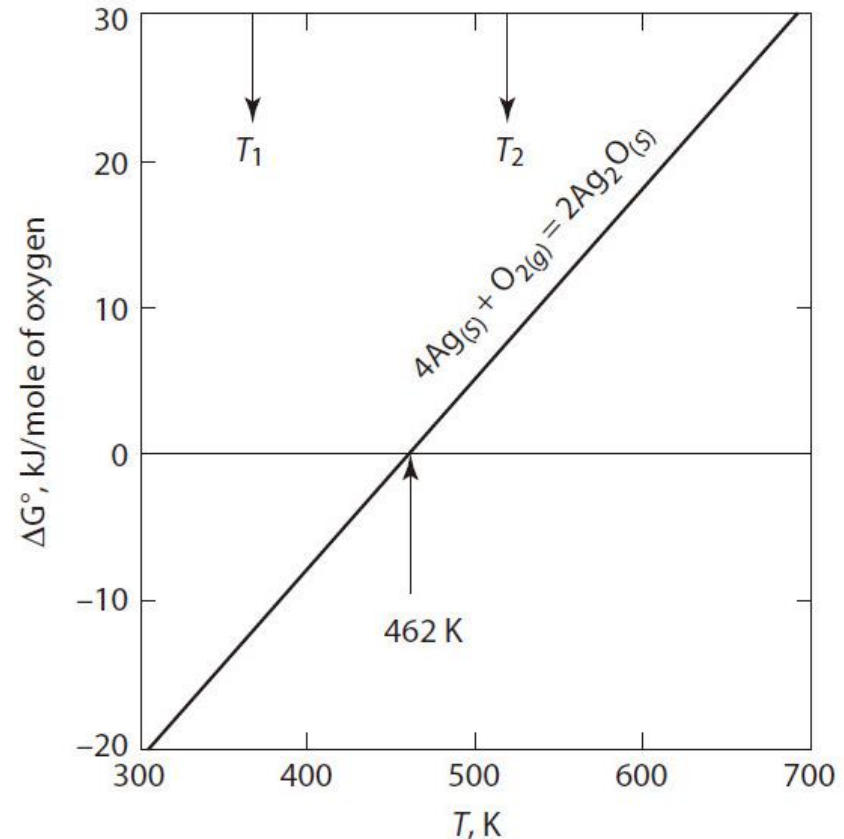


Figure 12.2 The Ellingham line for the oxidation of silver.

12.4 Ellingham Diagrams

- ✓ $\Delta G^\circ = 0$ at $T = 462$ K, and thus, at this temperature, pure solid silver and oxygen gas at 1 atm pressure are in equilibrium with pure solid silver oxide.
- ✓ From Equation 12.6, $\Delta G^\circ = -RT \ln K = RT \ln p_{\text{O}_2(\text{eq.}T)} = 0$ at 462 K, and, therefore, $p_{\text{O}_2(\text{eq.}462 \text{ K})} = 1$.
- ✓ If the temperature of the system (pure Ag_2O , pure $\text{Ag}_{(s)}$, and oxygen gas at 1 atm pressure) is decreased to T_1 , then, since ΔG° for the oxidation reaction becomes negative, the metal phase becomes unstable relative to silver oxide and oxygen gas at 1 atm pressure and is hence spontaneously oxidized.
- ✓ The value of $p_{\text{O}_2(\text{eq.}T_1)}$ is calculated from $\Delta G_{T_1}^\circ = RT_1 \ln p_{\text{O}_2(\text{eq.}T_1)}$, and, as $\Delta G_{T_1}^\circ$ is a negative quantity, $p_{\text{O}_2(\text{eq.}T_1)} < 1$ atm.

12.4 Ellingham Diagrams

- ✓ Similarly, if the temperature of the system is increased from 462 K to T_2 , then, since ΔG° for the oxidation becomes positive, the oxide phase becomes unstable relative to silver metal and oxygen gas at 1 atm and spontaneously dissociates.
- ✓ Since $\Delta G_{T_2}^\circ$ is a positive quantity, $p_{\text{O}_2(\text{eq.}T_2)}$ is greater than 1 atm. The value of ΔG° for oxidation is thus a measure of the chemical affinity of the metal for oxygen, and the more negative the value of ΔG° at any temperature, the more stable is the oxide.

12.4 Ellingham Diagrams

✓ For the oxidation reaction $A_{(s)} + O_{2(g)} = AO_{2(s)}$,

$$\Delta S^\circ = S_{AO_2(s)}^\circ - S_{O_2(g)}^\circ - S_{A(s)}^\circ$$

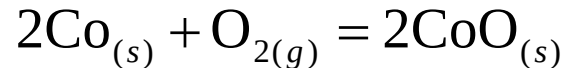
and since, generally, in the temperature range in which A and AO_2 are solid, $S_{O_2}^\circ$ is considerably greater than both S_A° and $S_{AO_2}^\circ$ (see, for example, Figure 6.15), then $\Delta S^\circ \approx -S_{O_2}^\circ$

12.4 Ellingham Diagrams

- ✓ Thus, the standard entropy changes for oxidation reactions involving solid phases have nearly the same values, which correspond to the disappearance of 1 mole of oxygen gas initially at 1 atm pressure.
- ✓ Since the slopes of the lines in an Ellingham diagram are equal to $-\Delta S^\circ$, the lines are more or less parallel to one another, as will be seen in the Ellingham diagram presented in Section 12.6.
- ✓ ΔG° at any temperature is the sum of the enthalpy contribution ΔH° (which is independent of T if $\Delta c_p = 0$) and the entropy contribution $-T\Delta S$ (which, if $\Delta c_p = 0$, is a linear function of temperature).

12.4 Ellingham Diagrams

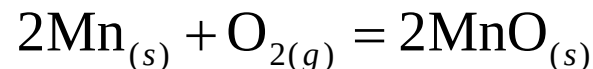
- ✓ The two contributions are illustrated in Figure 12.3 for the oxidation reaction of cobalt:



for which

$$\Delta G^\circ = -467,800 + 143.7T \text{ J}$$

in the temperature range 298–1763 K, and for the oxidation reaction of manganese,



for which

$$\Delta G^\circ = -769,400 + 145.6T \text{ J}$$

in the temperature range 298–1500 K.

12.4 Ellingham Diagrams

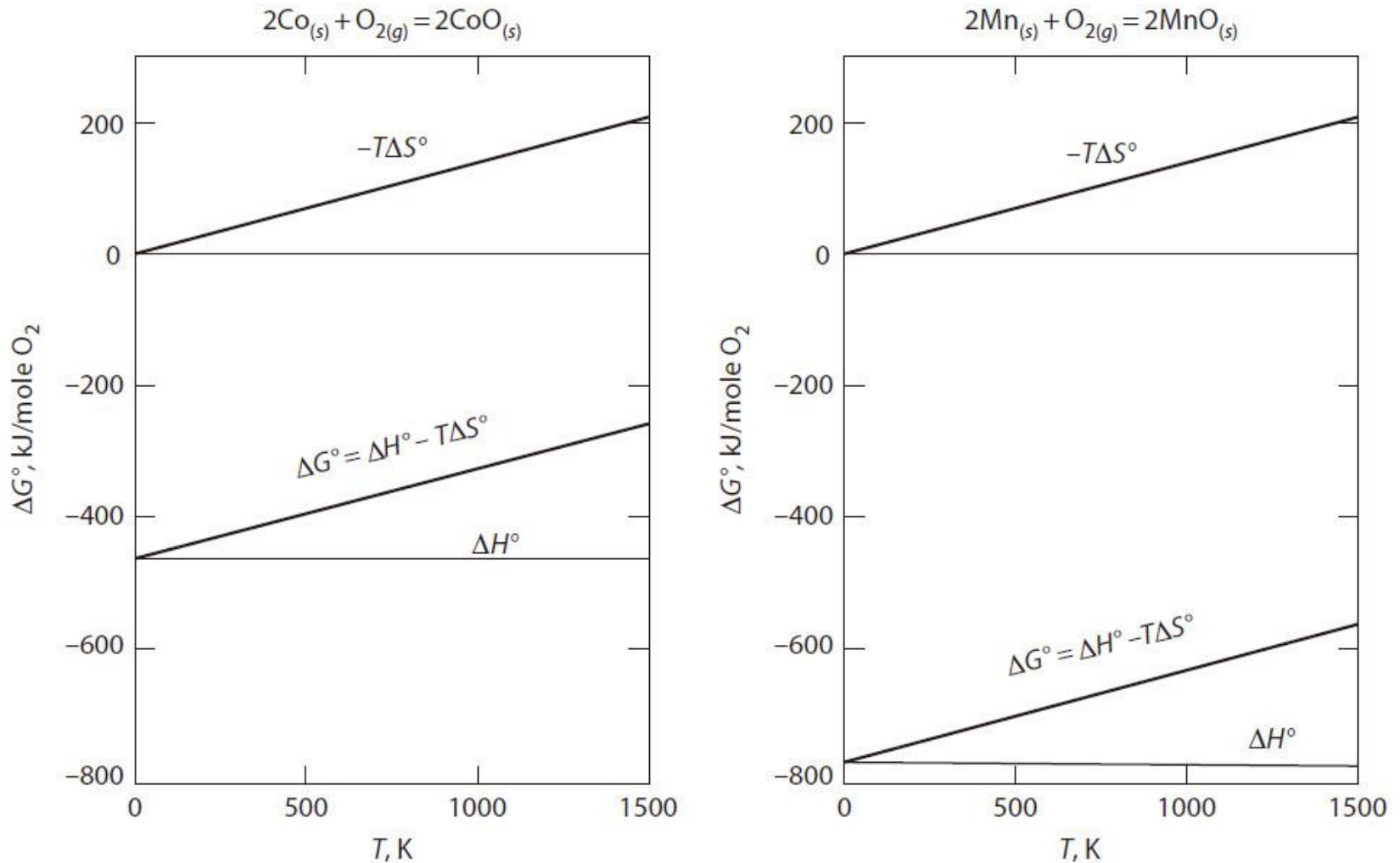


Figure 12.3 Illustration of the effect of the magnitude of ΔH° on the $\Delta G^\circ - T$ relationships for reactions of the type $2M_{(s)} + O_{2(g)} = 2MO_{(s)}$.

12.4 Ellingham Diagrams

- ✓ Since the values of ΔS° for these two reactions are virtually equal to one another, Figure 12.3 shows that the relative stabilities of the oxides CoO and MnO are determined by their values of ΔH° , in that the more negative the value of ΔH° , the more negative the value of ΔG° , and hence the more stable the oxide.

- ✓ Since

$$\ln K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} = \ln \left(\frac{1}{p_{\text{O}_2(\text{eq.}T)}} \right)$$

then

$$p_{\text{O}_2(\text{eq.}T)} = \exp \frac{\Delta H^\circ}{RT} \exp \frac{-\Delta S^\circ}{R} = \text{constant} \times \exp \frac{\Delta H^\circ}{RT}$$

- ✓ Since ΔH° is a negative quantity, $p_{\text{O}_2(\text{eq.}T)}$ increases exponentially with increasing temperature and, at any temperature, $p_{\text{O}_2(\text{eq.}T)}$ decreases as ΔH° becomes more negative.

12.4 Ellingham Diagrams

- ✓ Consider two oxidation reactions, the Ellingham lines of which intersect each other; for example,



and



which are shown in Figure 12.4.

- ✓ From Figure 12.4 it is seen that $\Delta H_{(ii)}^\circ$ is more negative than $\Delta H_{(i)}^\circ$ and that $\Delta S_{(ii)}^\circ$ is more negative than $\Delta S_{(i)}^\circ$.
- ✓ The subtraction of reaction (i) from reaction (ii) gives



for which the variation of ΔG° with T is as shown in Figure 12.5.

12.4 Ellingham Diagrams

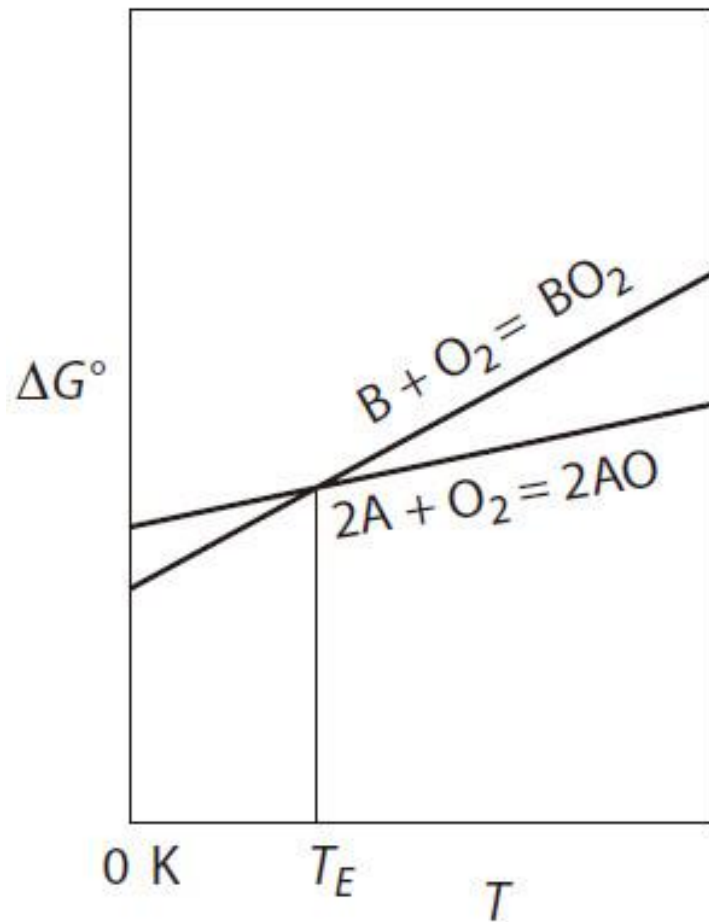


Figure 12.4 Intersecting Ellingham lines for two hypothetical oxidation reactions.

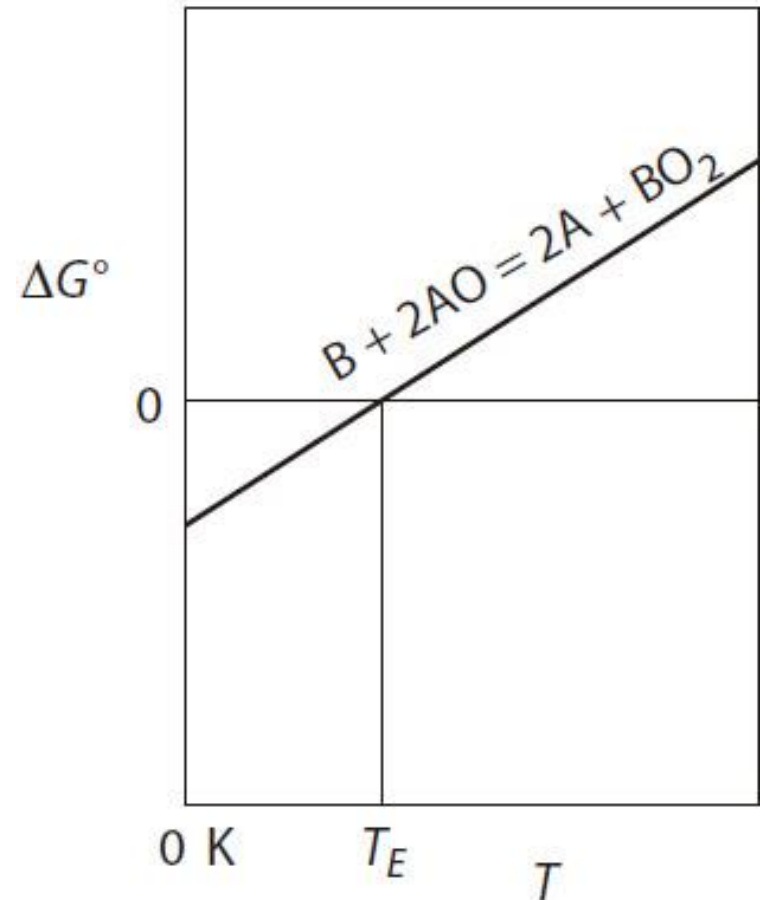


Figure 12.5 The variation of ΔG° with T for $B + 2AO = 2A + BO_2$ from Figure 12.4.

12.4 Ellingham Diagrams

- ✓ At temperatures less than T_E , A and BO_2 are stable with respect to B and AO, and at temperatures higher than T_E , the reverse is the case. At T_E , A, B, AO, and BO_2 , occurring in their standard states, are in equilibrium with one another. The equilibrium at T_E (as with any equilibrium) occurs as the result of a compromise between enthalpy and entropy considerations.
- ✓ Since $\Delta H_{(\text{iii})}^\circ = \Delta H_{(\text{ii})}^\circ - \Delta H_{(\text{i})}^\circ < 0$ and $\Delta S_{(\text{iii})}^\circ = \Delta S_{(\text{ii})}^\circ - \Delta S_{(\text{i})}^\circ < 0$, the system $\text{A} + \text{B} + \text{O}_2$ has
 - A minimum enthalpy when the system occurs as $\text{A} + \text{BO}_2$
 - A maximum entropy when the system occurs as $\text{B} + \text{AO}$

12.4 Ellingham Diagrams

- ✓ At T_E , $\Delta H_{(iii)}^\circ$ equals $-T_E \Delta S_{(iii)}^\circ$, and thus, $\Delta G_{(ii)}^\circ = 0$. At temperatures less than T_E , the enthalpy contribution to $\Delta G_{(iii)}^\circ$ outweighs the entropy contribution, and thus, $\Delta G_{(iii)}^\circ$ is negative and $A + BO_2$ is the stable state. At temperatures greater than T_E , the reverse is the case: $\Delta G_{(iii)}^\circ$ is positive and $B + AO$ is the stable state.

12.4 Ellingham Diagrams

- ✓ Figure 12.5 thus shows that if pure A were to be used as a reducing agent to reduce pure BO_2 , to form pure B and pure AO, then the reduction would have to be conducted at temperatures greater than T_E .
- ✓ The foregoing discussion also illustrates that, in order to compare the stabilities of different oxides, the Ellingham diagrams must be drawn for oxidation reactions involving the consumption of the same number of moles of oxygen.

12.4 Ellingham Diagrams

- ✓ Thus, the unit of ΔG° for the oxidation reaction must be one of energy (e.g., joules) per mole of oxygen.
- ✓ In order to avoid having to calculate the value of $p_{\text{O}_2(\text{eq.}T)}$ for any oxidation reaction, Richardson* added a nomographic scale to the Ellingham diagram. This scale is constructed as follows.

12.4 Ellingham Diagrams

- ✓ At any temperature T , the standard Gibbs free energy change for an oxidation reaction, ΔG_T° , is given by Equation 12.6 as $RT \ln p_{\text{O}_2(\text{eq.}T)}$.
- ✓ However, from Equation 8.7,

$$G = G^\circ + RT \ln P$$

ΔG_T° is seen to be numerically equal to the decrease in the Gibbs free energy of 1 mole of oxygen gas when its pressure is decreased from 1 atm to $p_{\text{O}_2(\text{eq.}T)}$ atm at the temperature T .

12.4 Ellingham Diagrams

- ✓ Consider the variation of ΔG with T in Equation 8.7. For a decrease in the pressure of 1 mole of ideal gas from 1 atm to P atm, ΔG versus T is a straight line with a slope of $R \ln P$, and, since $P < 1$, the line has a negative slope.
- ✓ Similarly, for an increase in the pressure of 1 mole of ideal gas from 1 to P atm, the variation of ΔG with T is linear with a positive slope of $R \ln P$.

12.4 Ellingham Diagrams

- ✓ Thus, a series of lines can be drawn for given pressure changes (from 1 to P atm) as a function of temperature. These lines radiate from the point $\Delta G = 0$, $T = 0$, as shown in Figure 12.6.

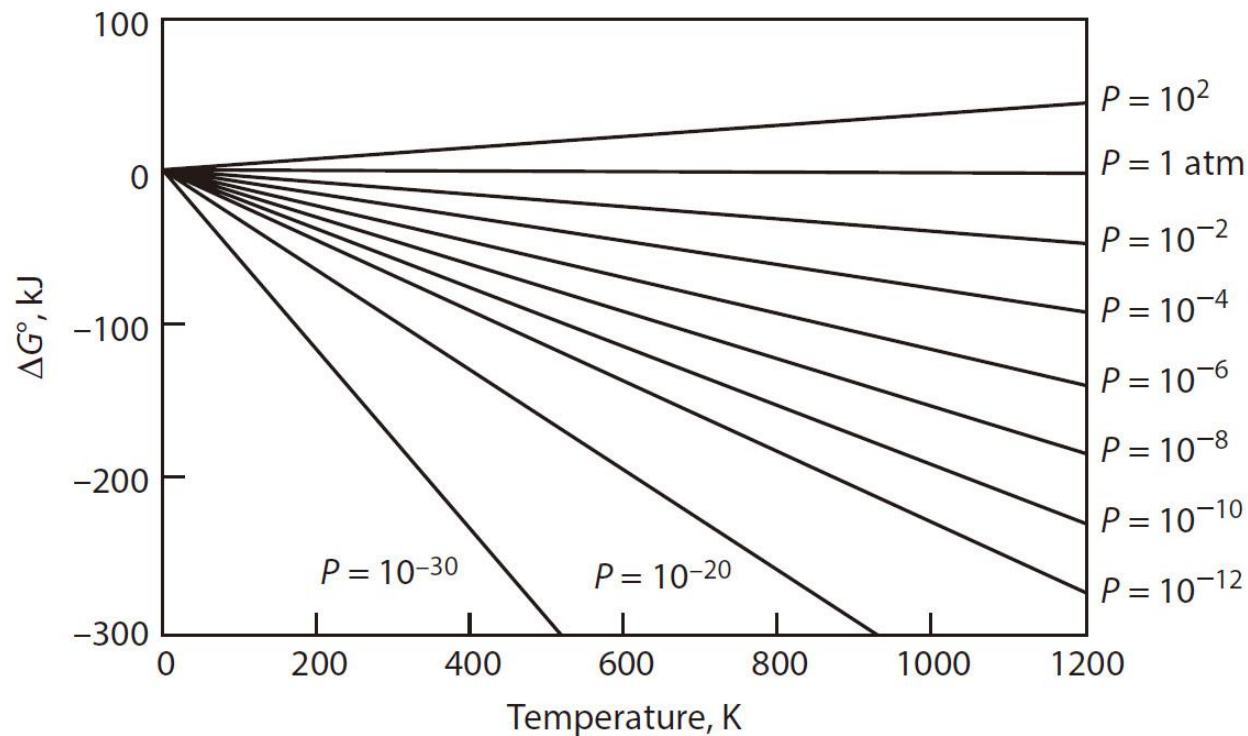


Figure 12.6 The variation, with temperature, of the difference between the Gibbs free energy of 1 mole of ideal gas in the state ($P = P \text{ atm}$, T) and the Gibbs free energy of 1 mole of ideal gas in the state ($P = 1 \text{ atm}$, T).

12.4 Ellingham Diagrams

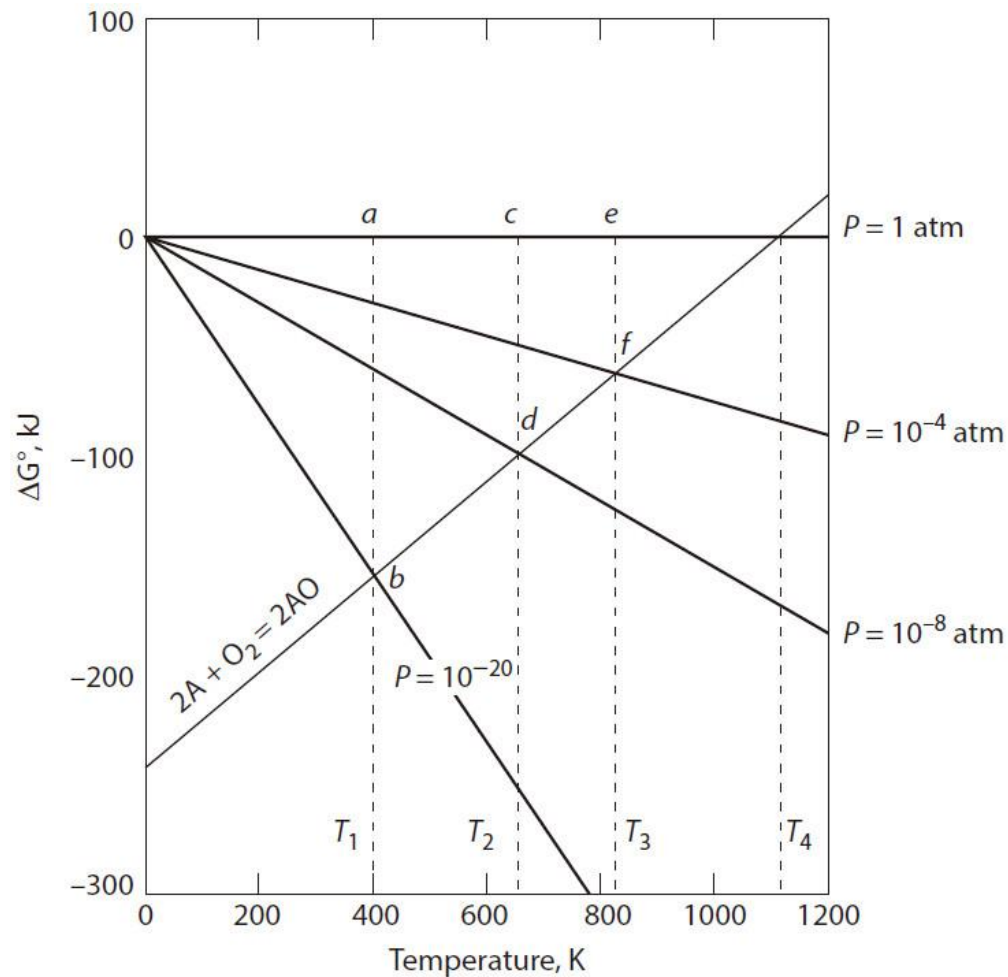


Figure 12.7 The superimposition of an Ellingham line onto Figure 12.6.

12.4 Ellingham Diagrams

- ✓ The superimposition of Figure 12.6 with a typical Ellingham diagram is shown in Figure 12.7. In Figure 12.7,
- At T_1 , $\Delta G_{T_1}^\circ = ab$ = the decrease in Gibbs free energy when p_{O_2} is decreased from 1 atm to 10^{-20} atm at T_1 .
 - At T_2 , $\Delta G_{T_2}^\circ = cd$ = the decrease in Gibbs free energy when p_{O_2} is decreased from 1 atm to 10^{-8} atm at T_2 .
 - At T_3 , $\Delta G_{T_3}^\circ = ef$ = the decrease in Gibbs free energy when p_{O_2} is decreased from 1 atm to 10^{-4} atm at T_3 .
 - At T_4 , $\Delta G = 0$, which corresponds to no change of p_{O_2} from 1 atm. Thus, $p_{\text{O}_2(\text{eq.}T_4)} = 1 \text{ atm}$.

12.4 Ellingham Diagrams

- ✓ The $p_{\text{O}_2(\text{eq.}T)}$ nomographic scale is thus added to the Ellingham diagram along the right-hand edge and along the bottom edge. The value of $p_{\text{O}_2(\text{eq.}T)}$ for any metal–metal oxide equilibrium is read off the graph as that value on the scale which is collinear with the points $\Delta G^\circ = 0$, $T = 0$, and ΔG_T° , $T = T$.
- ✓ The reactions (i) and (ii) shown in Figure 12.4 can be reexamined using the $p_{\text{O}_2(\text{eq.}T)}$ nomographic scale. Figure 12.4 is reproduced with a nomographic oxygen pressure scale in Figure 12.8.

12.4 Ellingham Diagrams

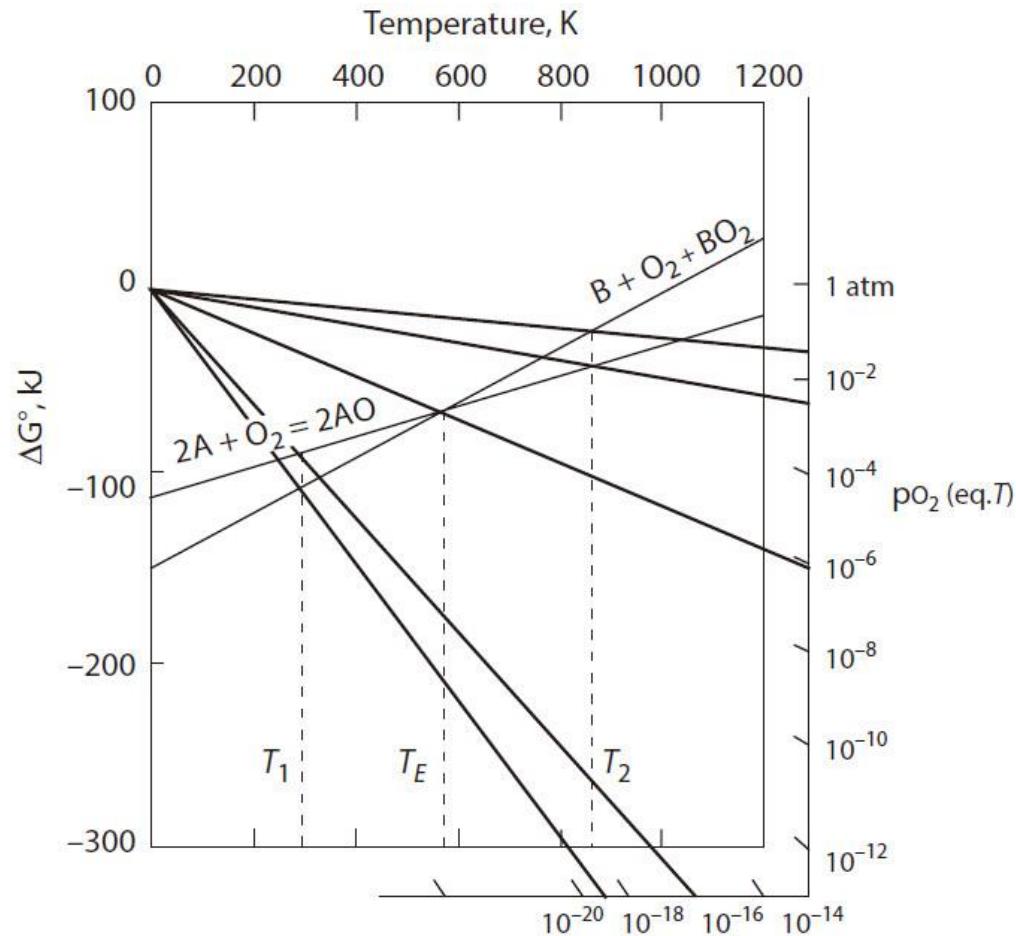


Figure 12.8 Illustration of the addition of the Richardson oxygen pressure nomographic scale to an Ellingham diagram.

12.4 Ellingham Diagrams

- ✓ At any temperature lower than T_E (say, T_1), it is seen that

$$p_{\text{O}_2}(\text{equation for reaction (ii) at } T_1) < p_{\text{O}_2}(\text{equation for reaction (i) at } T_1)$$

- ✓ Thus, if metal A and metal B are placed in a closed system in an atmosphere of oxygen at $P = 1$ atm, both metals spontaneously oxidize.

12.4 Ellingham Diagrams

- ✓ As a consequence of the consumption of oxygen to form the oxides, the pressure of oxygen decreases. Oxidation of both metals continues until the oxygen pressure is decreased to the value p_{O_2} (equation for reaction (i) at T_1), at which point the oxidation of A ceases.
- ✓ However, since $\text{B} + \text{O}_2$ at p_{O_2} (equation for reaction (i) at T_1) is still unstable with respect to BO_2 , the oxidation of B continues until the oxygen pressure has been decreased to p_{O_2} (equation for reaction (ii) at T_1).

12.4 Ellingham Diagrams

- ✓ Since the oxygen pressure is decreased below p_{O_2} (equation from reaction (i) at T_1), then AO becomes unstable with respect to A and O_2 at the prevailing pressure, and hence AO decomposes.
- ✓ When complete equilibrium is attained, the state of the system is A + $BO_2 + O_2$ at p_{O_2} (equation for reaction (ii) at T_1).

12.4 Ellingham Diagrams

- ✓ At any temperature higher than T_E (say, T_2),

$$p_{\text{O}_2}(\text{equation for reaction (i) at } T_2) < p_{\text{O}_2}(\text{equation for reaction (ii) at } T_2)$$

and an argument similar to this shows that the equilibrium state of the closed system containing, initially, $A + B + \text{O}_2$ at 1 atm pressure, is $B + \text{AO} + \text{O}_2$ at p_{O_2} (equation for reaction (i) at T_2).

- ✓ It is thus obvious that A , B , AO , BO_2 , and an oxygen atmosphere are in equilibrium only at that temperature T at which

$$p_{\text{O}_2(\text{eq.}[i]T)} = p_{\text{O}_2(\text{eq.}[ii]T)}$$

- ✓ Figure 12.8 shows that this unique temperature is T_E , the temperature at which the Ellingham lines intersect one another. The system has no degrees of freedom.

Homework

1. To what temperature must MgCO_3 be heated in an atmosphere containing a partial pressure of CO_2 of 10^{-2} atm to cause the decomposition of the carbonate?
2. Using the standard Gibbs free energies of formation of NiO from solid Ni and liquid Ni, calculate the melting temperature, molar heat of melting, and the molar entropy of melting of nickel.

Homework

3. Calculate the temperature at which pure Ag_2O decomposes to Ag metal and O_2 gas when heated in (a) pure oxygen at 1 atm pressure, and (b) air.
4. Determine the maximum pressure of water vapor in wet hydrogen at 1 atm pressure in which chromium can be heated without oxidation occurring at 1500 K. Is the oxidation of Cr by water vapor exothermic or endothermic?

Thermodynamics of Materials & Phase transformations

The End

